

Serial tutoring reveals a composite song template

Abstract

A popular animal model for vocal learning is the zebra finch. According to the standard theory (Konishi, 1965), a young zebra finch first stores a memory of a heard song, a so-called template, and later uses the template to refine its own vocal output until it matches the song it heard. Despite more than 50 years of evidence for template-driven song learning, it is not yet clear how birds extract this template from their auditory experience. Is the template simply a single song instance or does it derive from several instances? To understand the template formation process, we conducted serial tutoring experiments on juvenile zebra finches. We serially exposed juveniles to two different tutor playbacks, first a source song and later a target song. When the second syllable in the target deviates by one semitone from the corresponding syllable in the source ($AB \rightarrow AB^+$), juveniles ended up learning neither the source nor the target syllable, but a syllable in between, showing that a template (i.e., the song they end up learning) can be a composite generalizing over auditory memories. In another experiment where the target song contained two different variants of the corresponding source (one variant one semitone higher in pitch than the original, the other one semitone lower: $ABAB \rightarrow AB^+AB^-$), juveniles shifted their syllable to one variant and created a new syllable that matched the other variant, paradoxically learning a more complex auditory input better. To model this behavior, we hypothesized that birds have a built-in mechanism of generalizing over auditory memories. When forming a template, they first infer the causal structure underlying the heard songs and then they compute the optimal template, given the posterior of possible causal structures. This model is able to reproduce the templates extracted by juveniles in all experiment groups.

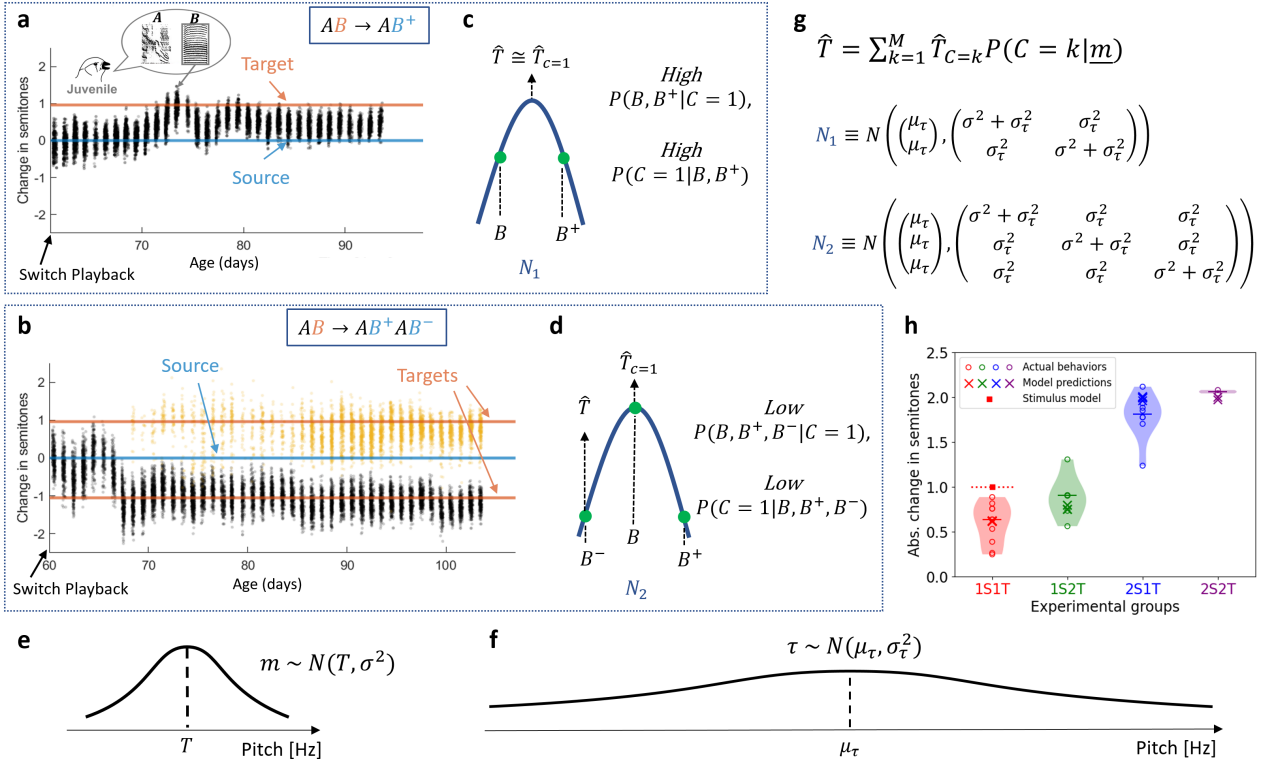
Additional detail

Behavioral data: We collected a total of 26 syllables from 19 juvenile zebra finches (**h**), divided into four groups depending on two factors: 1) how many semitones a target syllable was away from the corresponding source syllable, and 2) how many new target syllables were provided. In a one-target experiment, the harmonic syllable B from the source was either shifted up or down in pitch to form the target (shifted by either 1 or 2 semitones). In a two-targets experiment, the target contained both up- and down-shifted versions of the source syllable. The resulting four bird groups were: one-semitone one-target (1S1T), one-semitone two-targets (1S2T), two-semitones one-target (2S1T), and two-semitones two-targets (2S2T). The two-semitones birds were taken from a previous study (Lipkind et al., 2017).

We first segmented the juveniles' songs using WhisperSeg (Gu et al., 2023) and computed the pitch of vocal segments using the Praat autocorrelation algorithm (Boersma, 1993). Figures **a** and **b** show pitch scatter plots from a bird in a 1S1T experiment ($AB \rightarrow AB^+$) and from a bird in a 1S2T experiment ($AB \rightarrow AB^+AB^-$), respectively (each dot represents the pitch of a syllable; the y axis reports the pitch difference between a syllable and the source syllable; the blue line indicates the source syllable and the orange lines the target syllables; the plots show days following the switch to target). To match the second target in **b**, the bird recruits a call (orange dots), but this process is not to be further explored in this study.

In a previous study (Lipkind et al., 2017), it was found that in 2S1T and 2S2T experiments, juveniles shift the pitch of their syllable to (one of) the target(s). Figure **a** shows, however, that in a 1S1T experiment, the pitch does not converge to the target, but remains somewhere in between the source and the target, clashing with notions that a template is a single instance of an auditory memory. Therefore, we argue that a template is more generally a prototype composed of different auditory memories, rather than an exemplar. Figure **b** shows that in a 1S2T experiment, when the target is still one semitone away from the source, but a second target is introduced on the opposite side of the pitch axis, the juvenile's syllable gets closer to a target, either B^+ and B^- , than when the target is presented alone (1S1T case shown in **a**). Had the strategy for template formation been simple averaging, we would have expected the juvenile's syllable to remain at B , since B is the average of B, B^+ , and B^- .

Modeling: We assume that given a template, there is a bell-shaped likelihood function that an auditory memory is associated with: We view an auditory memory $m = B, B^+$, or B^- as a noisy measurement of a real template T under a Gaussian assumption, $m \sim N(T, \sigma^2)$, where σ^2 is the variance of auditory memories (**e**). We assume that when forming a template, a bird weighs the



posterior probability that auditory memories are caused by a common template, $P(C=1|B, B^+)$ in **c** and $P(C=1|B, B^+, B^-)$ in **d**, where $C=1$ denotes the event that auditory memories are caused by one common template. The Gaussians in **c** and **d** depict the likelihood functions (simplified as 1-dimensional Gaussians for the purpose of visualization), where $\hat{T}_{C=1}$ is the least-squares Bayes optimal template under $C=1$. In **c**, the inferred template $\hat{T}$ is close to $\hat{T}_{C=1}$ because a high likelihood $P(B, B^+|C=1)$ given a common template leads to a high posterior probability $P(C=1|B, B^+)$. In **d**, a low likelihood and therefore a low posterior probability leads to a $\hat{T}$ that is far away from $\hat{T}_{C=1}$.

Formally, we model template formation using Bayesian causal inference (Körding et al., 2007). The first equation in **g** shows how the template is inferred. The model takes as input M auditory memories m and returns as output the inferred template $\hat{T}$. The inferred template is a weighted sum of different Bayes optimal templates $\hat{T}_{C=k}$ for different causal structures ($C=k$; number of templates). The second and third equations are the analytical expressions for the likelihoods in **c** and **d**, assuming a template prior distribution $N(\mu_\tau, \sigma_\tau^2)$. A template prior is a zebra finch's innate preference towards regarding a certain memory as a template. Since $C=1$ does not assume a fixed template, (meaning that as long as a template is common, it does not matter what the actual template is) we marginalize over all possible templates (τ in **f**). Because zebra finches produce a wide frequency range, σ_τ^2 should ideally be large.

Setting the prior probabilities $P(C=k)$ to be uninformative priors of $1/M$, the model comprises of 3 free parameters: σ , μ_τ , and σ_τ . Figure **h** shows the result of fitting the model to the 26 syllables using SciPy's optimization tools (Virtanen et al., 2020). The circles and the violin plots show the actual behaviors, the crosses show the model predictions, and the red square shows the stimulus model prediction (explained in the next paragraph). The fitted parameters are as follows: $\sigma = 0.27$ semitones, $\mu_\tau = 1869\text{Hz}$, and $\sigma_\tau = 37.53$ semitones. The results are encouraging because not only is σ_τ large and μ_τ within the singing range of zebra finches, but also because σ is small. Studies have shown that zebra finches are excellent memorizers of songs and that they have the ability to discriminate small auditory differences (Narula et al., 2021, Elie et al., 2021), suggesting a small auditory memory variance σ^2 .

Lastly, we demonstrate the importance of using memories, not auditory stimuli, as model inputs. In 1S1T experiments, juveniles' template lies in between the two playbacks on average (**h**, solid red bar). If the template were formed directly from auditory stimuli (**h**, dotted red bar), there would be a factor continually weighing $\hat{T}$ towards the stimuli. As a result, the template would converge to the target (**h**, red square). In contrast, a template model built with memories is able to decouple the effect of hearing an individual song on the template, meaning that a heard song may strengthen an auditory memory but does not by itself impact a template. Therefore, in a memory-based model (**h**, crosses), the template remains a stable goal for the bird's vocal output to match.